Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Mia rides 1/2 mile in 1/4 hour. What is her unit rate in miles per hour? Student's answer: 1/8

Divide the distance by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 1/2 by 1/4. Divide: 1/2 ÷ 1/4 = 2 miles per hour.	**2 miles per hour**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3/4 pound of cheese costs $6. What is the price per pound? Student's answer: $4.50

Divide the cost by the weight.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied $6 by 3/4. Divide: $6 ÷ 3/4 = $8 per pound.	**$8 per pound**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Mario walks 1 1/2 miles in 1/3 hour. What is his speed in miles per hour? Student's answer: 1/2 mile per hour

Divide the distance by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the wrong way. 1 1/2 ÷ 1/3 = 4 1/2 miles per hour.	**4 1/2 miles per hour**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Bella types 150 words in 2 1/2 minutes. How many words per minute? Student's answer: 375

Divide the words by the minutes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 150 by 2 1/2. Divide: 150 ÷ 2.5 = 60 words per minute.	**60 words per minute**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A snail moves 3/5 foot in 1/4 minute. How many feet per minute? Student's answer: 3/20

Divide the distance by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the fractions. Divide: 3/5 ÷ 1/4 = 12/5 = 2 2/5 feet per minute.	**2 2/5 feet per minute**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is faster: 4 miles in 1/2 hour or 7 miles in 1 hour? Student says: 7 miles in 1 hour

Find the speed of each.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
4 miles in 1/2 hour is 8 miles per hour. That is faster than 7 miles per hour.	**4 miles in 1/2 hour**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is this table proportional? x = 1, 2, 3, 4 and y = 3, 6, 9, 10. Student says: Yes

Check whether y/x is the same for every pair.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
For (4, 10), y/x = 2.5, but the other pairs give 3. The table is not proportional.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

4 items cost $12 in total. What is the constant of proportionality for cost per item? Student's answer: 4

Divide the cost by the number of items.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the number of items. The constant is $12 ÷ 4 = $3 per item.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The equation is y = 2.5x. What is y when x = 8? Student's answer: 10.5

Substitute 8 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 2.5 and 8. Multiply: 2.5 × 8 = 20.	**20**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $y = 4x + 1$ a proportional relationship? Student says: Yes

Check whether the line passes through (0, 0).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The +1 means y is 1 when x is 0. A proportional relationship always passes through (0, 0), so this one is not proportional.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table shows (2, 5) and (4, 10). What is the constant of proportionality? Student's answer: 2

Divide y by x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the x-value. Divide y by x: 5 ÷ 2 = 2.5.	**2.5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Sales commission is 10% of sales. If sales are $250, what is the commission? Student's answer: $2.50

Write 10% as a decimal and multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided by 100 instead of 10. 0.10 × $250 = $25.	**$25**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A restaurant bill is $40. What is a 15% tip? Student's answer: $15

Find 15% of $40.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found 37.5%, not 15%. 0.15 × $40 = $6.	**$6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An $80 item increases in price by 25%. What is the new price? Student's answer: $105

Find 25% of $80, then add it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 25% of the wrong amount. 25% of $80 is $20, so the new price is $100.	**$100**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A $25 book has an 8% sales tax. What is the total price? Student's answer: $2

Find the tax, then add it to the price.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2 is the tax alone. Add it to the price: $25 + $2 = $27.	**$27**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

18 of 24 students passed a test. What percent passed? Student's answer: 25%

Write the fraction and change it to a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found the percent who did not pass. 18/24 = 75% passed.	**75%**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A price drops from $60 to $45. What is the percent decrease? Student's answer: 15%

Find the change, then divide by the original price.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the amount of the change, not the percent. The decrease is $15, and 15 ÷ 60 = 25%.	**25%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A $200 item is marked 5% off. What is the sale price? Student's answer: $10

Find the discount, then subtract it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$10 is the discount. The sale price is $200 - $10 = $190.	**$190**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add -3 + 8. Student's answer: -11

Find the difference of the absolute values, then use the sign of the larger one.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the absolute values. 8 is larger and positive, so -3 + 8 = 5.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract 4 - (-6). Student's answer: -2

Subtracting a negative means adding its opposite.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 6. Subtracting -6 is the same as adding 6: 4 + 6 = 10.	**10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add -2.5 + (-4). Student's answer: 6.5

Add the absolute values and keep the sign.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sum of two negatives is negative. -2.5 + (-4) = -6.5.	**-6.5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 3/4 + (-1/2). Student's answer: -1/4

Use a common denominator.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The positive number is larger, so the answer is positive. 3/4 - 1/2 = 1/4.	**1/4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Subtract -7 - (-3). Student's answer: -10

Subtracting a negative means adding its opposite.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Subtracting -3 is the same as adding 3: -7 + 3 = -4.	**-4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The temperature is -8°F and rises 13 degrees. What is the new temperature? Student's answer: -21°F

Start at -8 and move up 13.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Rising means adding. -8 + 13 = 5°F. The student subtracted.	**5°F**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply (-4) × (-6). Student's answer: -24

Use the sign rules.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Two negatives multiply to a positive. (-4) × (-6) = 24.	**24**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide (-12) ÷ 3. Student's answer: 4

Use the sign rules.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A negative divided by a positive is negative. -12 ÷ 3 = -4.	**-4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply (-2/3) × (3/4). Student's answer: 1/2

Multiply the numerators and denominators.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The product of a negative and a positive is negative. The answer is -6/12 = -1/2.	**-1/2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide $-15 \div (-5/2)$. Student's answer: -6

Multiply by the reciprocal, then apply the sign rules.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Two negatives make a positive. $-15 \times (-2/5) = 6$.	**6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply 0.5 × (−0.4). Student's answer: 0.2

Multiply the numbers, then check the sign.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A positive times a negative is negative. 0.5 × 0.4 = 0.2, so the answer is -0.2.	**-0.2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide (-3/4) ÷ (1/2). Student's answer: -3/8

Multiply by the reciprocal of 1/2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 1/4 instead of 2. (-3/4) × 2 = -3/2.	**-3/2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A team gains 3 points, then loses 5 points. What is the net change? Student's answer: 2

Write the gain as positive and the loss as negative.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The loss is negative. 3 + (-5) = -2 points.	**-2 points**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The temperature falls from -4°F to -12°F. What is the change in temperature? Student's answer: 8°F

Subtract the starting value from the ending value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The temperature went down, so the change is negative. -12 - (-4) = -8°F.	**-8°F**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A fish sinks 2.5 feet every minute for 4 minutes. What is its position change in feet? Student's answer: 10 feet

Multiply the rate by the time and use a negative sign for sinking.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Sinking is a negative change. 2.5 × 4 = 10, so the change is -10 feet.	**-10 feet**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

count gets $45 in deposits and $62.50 in withdrawals. What is the change in the balance? Student's answe

Write the withdrawals as negative amounts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The balance went down. 45 - 62.50 = -17.50.	**-$17.50**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

liver starts at -30 feet, rises 12 feet, then descends 20 feet. What is the final position? Student's answer: -18 f

Follow each move in order.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student stopped after the rise. Then -18 - 20 = -38 feet.	**-38 feet**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

arns $1.25 per item for 40 items and has costs of $60. Is it a profit or loss, and how much? Student's answer:

Write the earnings as positive and the costs as negative.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Earnings are $50 and costs are $60. 50 - 60 = -10, which is a loss of $10.	**A loss of $10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $3(x + 2) - x$. Student writes: $2x + 2$

Distribute, then combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 must multiply the 2 as well: $3x + 6$. Then $3x + 6 - x = 2x + 6$.	**$2x + 6$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify 2x + 5 - 4x + 1. Student writes: -2x + 4

Combine like terms, keeping the signs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The constant terms are 5 + 1 = 6, not 4. The answer is -2x + 6.	**-2x + 6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $4(2y - 1) + 3$. Student writes: $8y + 3$

Distribute the 4 to both terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$4 \times (-1) = -4$, not $+4$ or 0. So $8y - 4 + 3 = 8y - 1$.	**$8y - 1$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify (1/2)x + (1/2)x. Student writes: x/2

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Two halves of x make one whole x. The answer is x.	**x**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify -(3a - 2). Student writes: -3a - 2

Distribute the negative sign to both terms.

What did the student do wrong? Why do you think the student made this error?

The negative sign changes the sign of the -2 too. -(3a - 2) = -3a + 2.

Solve the problem correctly. Show all of your work.

-3a + 2

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify 0.5(4m) - 1.5m. Student writes: 3.5m

Multiply first, then combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
0.5 × 4m = 2m, and 2m - 1.5m = 0.5m. The student added instead of subtracting.	**0.5m**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 4x + 10 in factored form. Student writes: 4(x + 2.5)

Find the greatest common factor of the coefficients.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The factor 2.5 is not an integer. The GCF of 4 and 10 is 2, so the answer is 2(2x + 5).	**2(2x + 5)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

An item is 15% off a price p. Which expression shows the sale price? Student writes: 0.15p

Think about what part is paid.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
0.15p is the discount. The price after 15% off is 0.85p.	**0.85p**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Factor 9y - 6. Student writes: 3(3y + 2)

Factor out the greatest common factor and check the signs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sign of the 2 is wrong. 3(3y + 2) = 9y + 6. The correct factored form is 3(3y - 2).	**3(3y - 2)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 0.4x + 0.6x equal to x? Student says: No

Combine like terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
0.4 + 0.6 = 1, so 0.4x + 0.6x = x. The expressions are equal.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Tim says $5(2 + x) = 10 + x$. Is he correct? Student says: Yes

Distribute the 5 to both terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The 5 multiplies x too. $5(2 + x) = 10 + 5x$, not $10 + x$.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 12m + 18 in factored form. Student writes: 3(4m + 6)

Find the greatest common factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 is a factor, but it is not the greatest. The GCF of 12 and 18 is 6, so the answer is 6(2m + 3).	**6(2m + 3)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Estimate 4.8 × 6.2. Student's estimate: 60

Round each number to a nearby whole number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Round 4.8 to 5 and 6.2 to 6. 5 × 6 = 30, so the estimate is about 30.	**About 30**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Four notebooks cost $6.75 each, plus $3 for tax. What is the total? Student's answer: $24

Multiply, then add the tax.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted the tax. 4 × $6.75 = $27, and $27 + $3 = $30.	**$30**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2x + 3 = 11$. Student's answer: 8

Undo the addition, then undo the multiplication.

What did the student do wrong? Why do you think the student made this error?

The student stopped after subtracting 3. Divide by 2: x = 4.

Solve the problem correctly. Show all of your work.

4

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A shirt costs $18 and the tax rate is 6%. What is the total price? Student's answer: $18.08

Find 6% of $18 and add it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
6% of $18 is $1.08, not $0.08. The total is $19.08.	**$19.08**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 0.5x - 2 = 5. Student's answer: 7

Undo the subtraction, then the multiplication.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student stopped too soon. Add 2 to get 0.5x = 7, then divide by 0.5 to get x = 14.	**14**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3(x - 2) = 12$. Student's answer: 2

Divide both sides by 3, then solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$12 \div 3 = 4$, so $x - 2 = 4$. Add 2 to get $x = 6$. **The student subtracted 2 from 4 instead.**	**6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $4x + 5 = 25$. Student's answer: 7.5

Subtract 5 from both sides, then divide by 4.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 5 to 25. Subtract 5: $4x = 20$, so $x = 5$.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3(x + 1) = 18$. Student's answer: 6

Divide by 3, then solve.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
18 ÷ 3 = 6, so x + 1 = 6. Subtract 1 to get x = 5.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2x - 7 = 9$. Student's answer: 16

Add 7 to both sides, then divide by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found $2x = 16$ and stopped. Divide by 2: $x = 8$.	**8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the inequality $2x < 10$. Student writes: $x > 5$

Divide both sides by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Dividing by a positive number keeps the direction. The solution is $x < 5$.	**$x < 5$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the inequality $-3x \geq 12$. Student writes: $x \geq -4$

Divide both sides by -3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Dividing by a negative number reverses the inequality sign. The solution is $x \leq -4$.	**$x \leq -4$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the inequality $x/2 + 1 > 4$. Student writes: $x > 3$

Undo the addition, then the division.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not multiply by 2. Subtract 1 to get $x/2 > 3$, then multiply by 2: $x > 6$.	**$x > 6$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

p scale is 1 cm = 5 km. Two cities are 3.5 cm apart on the map. How far apart are they? Student's answer: 3

Multiply the map distance by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not use the scale. 3.5 × 5 = 17.5 km.	**17.5 km**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ving has a scale of 1 inch = 4 feet. A room is 12 feet long. How long is it in the drawing? Student's answer: 48

Divide the actual length by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied 12 by 4. Divide: 12 ÷ 4 = 3 inches.	**3 inches**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A scale drawing uses 1 cm = 2 m. A wall is 6 m long. How long is it in the drawing? Student's answer: 12 cm

Divide the actual length by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied instead of dividing. 6 ÷ 2 = 3 cm.	**3 cm**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ap scale is 1 inch = 10 miles. Two towns are 2.5 inches apart. How far apart are they? Student's answer: 4 m

Multiply the map distance by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided 10 by 2.5. Multiply: 2.5 × 10 = 25 miles.	**25 miles**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A photo that is 4 inches wide is enlarged to 10 inches wide. What is the scale factor? Student's answer: 0.4

Divide the new width by the original width.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided the wrong way. 10 ÷ 4 = 2.5, so the scale factor is 2.5.	**2.5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s a scale of 1 inch = 8 feet. A line on the drawing is 2.25 inches long. What is the actual length? Student's ans

Multiply the drawing length by the scale.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not use the scale. 2.25 × 8 = 18 feet.	**18 feet**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have sides of 3 cm, 4 cm, and 5 cm? Student says: No

Check the triangle inequality.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3 + 4 is greater than 5, so these sides make a triangle. It is a right triangle.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have sides of 2 cm, 3 cm, and 6 cm? Student says: Yes

Check whether the two shorter sides add up to more than the longest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2 + 3 = 5, which is less than 6. These sides cannot form a triangle.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have angles of 60º, 60º, and 60º? Student says: No

Add the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
60 + 60 + 60 = 180, so the angles can form a triangle. It is an equilateral triangle.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have angles of 90°, 50°, and 50°? Student says: Yes

Add the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
90 + 50 + 50 = 190, which is more than 180. These angles cannot form a triangle.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have angles of 40°, 60°, and 80°? Student says: No

Add the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
40 + 60 + 80 = 180, so these angles form a triangle.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Can a triangle have sides of 1 cm, 2 cm, and 3 cm? Student says: Yes

Check whether the two shorter sides add up to more than the longest.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
1 + 2 = 3, which is not greater than 3. These sides cannot form a triangle.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lane cuts a cube parallel to one of its faces. What is the shape of the cross-section? Student's answer: triar

Picture the slice.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A slice parallel to a face has the same shape as that face. The cross-section is a square.	**Square**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

lane cuts a cylinder parallel to its bases. What is the shape of the cross-section? Student's answer: rectan

Picture the slice.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A slice parallel to the base has the same shape as the base. The cross-section is a circle.	**Circle**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ne cuts through two opposite edges of a cube. What is the shape of the cross-section? Student's answer: tr

Picture the slice.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The slice passes through two opposite edges and crosses the cube as a rectangle.	**Rectangle**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What shapes are in the net of a square pyramid? Student's answer: 4 squares and 1 triangle

List each face.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The base is a square, and the 4 sides are triangles. The net has 1 square and 4 triangles.	**1 square and 4 triangles**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many edges does a rectangular prism have? Student's answer: 8

Count the edges of the prism.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the vertices. A rectangular prism has 12 edges: 4 on the top, 4 on the bottom, and 4 vertical edges.	**12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s through the tip of a cone and cuts straight down through its center. What is the cross-section? Student's

Picture the slice.

What did the student do wrong? Why do you think the student made this error?

A vertical slice through the tip gives a triangle. A circle would come from a horizontal slice.

Solve the problem correctly. Show all of your work.

Triangle

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the circumference of a circle with radius 3 cm. Student's answer: 9π cm

Use C = 2πr.

What did the student do wrong? Why do you think the student made this error?

The student multiplied 3 by 3. C = 2π × 3 = 6π, which is about 18.85 cm.

Solve the problem correctly. Show all of your work.

6π cm, about 18.85 cm

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a circle with diameter 10 cm. Student's answer: 100π cm²

Find the radius first, then use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?

The student used the diameter in the area formula. The radius is 5, so $A = \pi \times 5^2 = 25\pi$ cm².

Solve the problem correctly. Show all of your work.

25π cm², about 78.5 cm²

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a circle with radius 5 cm. Student's answer: 10π cm²

Use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student doubled the radius. $A = \pi \times 5^2 = 25\pi$ cm².	**25π cm²**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A circle has a circumference of 12π cm. What is its diameter? Student's answer: 6 cm

Use C = πd.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the radius. C = πd, so d = 12π ÷ π = 12 cm.	**12 cm**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a circle with radius 2 cm. Student's answer: 2π cm²

Use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found the circumference. $A = \pi \times 2^2 = 4\pi$, which is about 12.57 cm².	**4π cm², about 12.57 cm²**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a half-circle with a diameter of 8 cm. Student's answer: 16π cm²

Find the radius, find the full area, then take half.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
16π is the area of the full circle with radius 4. A half-circle has an area of 8π cm².	**8π cm²**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are supplementary. One is 65°. What is the other? Student's answer: 25°

Supplementary angles add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 90°, which is for complementary angles. 180 - 65 = 115°.	**115°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are complementary. One is 35°. What is the other? Student's answer: 145°

Complementary angles add to 90°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 180°. 90 - 35 = 55°.	**55°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are vertical angles. One is 70°. What is the other? Student's answer: 110°

Vertical angles are equal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Vertical angles are equal. The other angle is 70°.	**70°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles of 45° and 60°. What is the third angle? Student's answer: 85°

Angles in a triangle add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted incorrectly. 180 − (45 + 60) = 180 − 105 = 75°.	**75°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles are complementary. One is x and the other is 40°. Student says x = 140°

Complementary angles add to 90°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 180°. x + 40 = 90, so x = 50°.	**50°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ngle of a triangle is 110°. One remote interior angle is 50°. What is the other remote interior angle? Student's

An exterior angle equals the sum of the two remote interior angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 110 and 50. The other remote angle is 110 - 50 = 60°.	**60°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has a base of 8 cm and a height of 5 cm. What is its area? Student's answer: 40 cm²

Use A = 1/2 bh.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot the 1/2. A = 1/2 × 8 × 5 = 20 cm².	**20 cm²**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A circular table has a radius of 3 feet. What is its area, to the nearest hundredth? Student's answer: 18.85 ft²

Use $A = \pi r^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
18.85 is the circumference. The area is $\pi \times 3^2 = 9\pi$, which is about 28.27 ft².	**About 28.27 ft²**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

m has a base area of 12 square units and a height of 5 units. What is its volume? Student's answer: 17 cubic

Use V = Bh.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 12 and 5. Multiply: 12 × 5 = 60 cubic units.	**60 cubic units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

r prism has a base area of 6 square units and a length of 10 units. What is its volume? Student's answer: 16

Use V = Bh.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 6 and 10. Multiply: 6 × 10 = 60 cubic units.	**60 cubic units**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box has a volume of 96 cubic cm. Its base is 8 cm by 4 cm. What is its height? Student's answer: 4 cm

Find the base area, then divide the volume by it.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided by 24, not 32. The base area is 8 × 4 = 32 square cm, so the height is 96 ÷ 32 = 3 cm.	**3 cm**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cylinder has a radius of 2 cm and a height of 5 cm. What is its volume? Student's answer: 10π cm³

Use $V = \pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to square the radius. $V = \pi \times 2^2 \times 5 = 20\pi$, which is about 62.83 cm³.	**20π cm³, about 62.83 cm³**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ers on the basketball team their favorite sport to learn about the whole school. Is this sample representativ

Think about who is in the sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sample is only basketball players, so it is not representative of the whole school.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

om sample of 50 students is taken from a school of 500. Is the sample likely to be representative? Student s

Think about how the sample was chosen.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A random sample is usually representative of the population, so this sample is likely representative.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

first 15 people in line at an ice cream shop about flavors for all the students. Is this sample representative?

Think about who is in the sample.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sample is not random, and it includes only people at the shop. It is not representative of all students.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What makes a sample representative of a population? Student's answer: It is a large sample.

Think about how the sample is chosen.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Size helps, but a sample is representative when it is chosen randomly from the population. A large biased sample is still biased.	**It is chosen randomly from the population**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

tudents from a school of 400 found 10 prefer blue. Estimate how many of all 400 students prefer blue. Stud

Find the proportion in the sample, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not scale up. 10 out of 40 is 1/4, and 1/4 of 400 is 100.	**100**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

eople from a town of 1,200 shows 15 people like pizza best. Estimate how many people in the town like pizza

Find the proportion in the sample, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
15 out of 60 is 1/4. Scaling to 1,200 gives 300 people.	**300**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ts from a school of 800, 20 students said they like the new lunch menu. Estimate how many students in the

Find the proportion, then scale up to the population.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the sample count. 20/50 = 40%, and 40% of 800 = 320.	**320**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

e of 100 students, 60 said yes to a question. Estimate the number of yes answers in a population of 1,500. St

Find the proportion, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student repeated the sample count. 60/100 = 60%, and 60% of 1,500 = 900.	**900**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

of 25 students has 8 who like soccer. What is the estimated percent of the population who like soccer? Stu

Write the proportion as a percent.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
8 out of 25 is 32%, not 8%.	**32%**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

200 from a city of 2,000 found 40 with brown eyes. Estimate the number of brown-eyed people in the city.

Find the proportion, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
40 out of 200 is 20%. Scaled to 2,000 it is 400 people.	**400**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

mple shows 60% prefer option A. Can we say exactly 60% of the population prefers option A? Student says:

Think about the variability in samples.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Samples vary. 60% is an estimate, and the true percent may be a little higher or lower.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

of 80 students from a school of 800 shows 24 read every day. Estimate the number who read every day. St

Find the proportion, then scale up.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
24 out of 80 is 30%. Scaled to 800 it is 240 students.	**240**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

up A has data 10, 12, 14. Group B has data 11, 13, 15. Which group has the higher mean? Student's answer: Grou

Find the mean of each group.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Group A's mean is 12, and Group B's mean is 13. Group B has the higher mean.	**Group B**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

up A has a range of 3 and Group B has a range of 10. Which group is more variable? Student's answer: Grou

Compare the spreads.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A larger range means more variability. Group B is more variable.	**Group B**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

as a median of 22 and Group B has a median of 33. What is the difference between the medians? Student's

Subtract the medians.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted the wrong numbers. 33 - 22 = 11.	**11**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

hean 5 and group B has the mean 9. Both groups have a range of 2. Is the difference in means meaningful

Compare the difference in means to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The difference of 4 is large compared to the range of 2. The difference is meaningful.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Set 1 is 5, 5, 5. Set 2 is 1, 5, 9. Which set has more spread? Student's answer: Set 1

Compare how far values are from each other.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Set 1 has no spread. Set 2 has values 1 and 9, so it has more spread.	**Set 2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

mean of 50. Group A has a range of 5 and group B has a range of 25. Which group is less consistent? Stude

Compare the ranges.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A larger range means less consistent. Group B has the larger range, so it is less consistent.	**Group B**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

deviation of 2. Group Y has a mean of 15 and a mean absolute deviation of 2. Is the difference in means lar

Compare the difference of means to the spread.

What did the student do wrong? Why do you think the student made this error?

The means differ by 3, which is larger than the spread of 2. The difference is meaningful.

Solve the problem correctly. Show all of your work.

Yes

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ool A has a mean of 7 hours of sleep. A random sample from School B has a mean of 7 hours. Student says:

Compare the means of the two samples.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The means are the same, so there is no difference to infer from these samples.	**No difference shown**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ndom samples have medians of 40 and 42, with spread of 15 in each. Student says: The difference is mear

Compare the difference to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A difference of 2 is small compared to a spread of 15. The difference is not meaningful.	**Not meaningful**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

test score of 78 and class 2 has an average of 80. The spread in each class is about 20. Student says: Class

Think about the spread and the sample.

What did the student do wrong? Why do you think the student made this error?

The difference of 2 is small compared to the spread of 20, so we cannot say for certain.

Solve the problem correctly. Show all of your work.

No, the difference is small

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

as a mean of 30 and a spread of 4. Group B has a mean of 36 and a spread of 4. Student says: The groups

Compare the difference to the spread.

What did the student do wrong? Why do you think the student made this error?

The difference of 6 is larger than the spread of 4, so the difference is meaningful.

Solve the problem correctly. Show all of your work.

Yes, the difference is meaningful

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

nean of 100 with a spread of 50. A sample from population 2 has a mean of 110 with a spread of 50. Student

Compare the difference to the spread.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The difference of 10 is small compared to the spread of 50. It is not clear that population 2 is higher.	**Not clearly higher**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A weather app says the chance of rain is 1.5. Can this be correct? Student says: Yes

Think about the range of probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Probabilities must be between 0 and 1. A value of 1.5 is not possible.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the probability of an impossible event? Student's answer: 1

Think about the range of probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
An impossible event has probability 0. An event that is certain has probability 1.	**0**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the probability of rolling a number less than 7 on a standard die? Student's answer: 0

Think about whether the event always happens.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Every roll is less than 7, so the event is certain. Its probability is 1.	**1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the probability of drawing a red card from a deck with 26 red cards out of 52? Student's answer: 2

Write the probability as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student wrote the count of cards, not a probability. The probability is 26/52 = 1/2.	**1/2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Letters A, B, C, and D are placed in a bag. What is the probability of picking a vowel? Student's answer: 1/2

Count the vowels out of the total letters.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Only A is a vowel, out of 4 letters. The probability is 1/4.	**1/4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is most likely: an event with probability 0.2, 0.7, or 0.45? Student's answer: 0.2

Compare the probabilities.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The largest probability is the most likely. 0.7 is the largest.	**0.7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s flipped 50 times and lands heads 28 times. What is the experimental probability of heads? Student's answ

Divide heads by total flips.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not divide by the total. 28/50 = 0.56.	**0.56**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

her is spun 100 times and lands on red 20 times. What is the experimental probability of red? Student's answ

Divide red outcomes by total spins.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the count. The probability is 20/100 = 0.2.	**0.2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

heoretical probability of heads of 0.5. A student flips it 10 times and gets 3 heads. Student says: The coin mu

Think about small samples.

What did the student do wrong? Why do you think the student made this error?

Experimental results vary, especially with few trials. 10 flips is too few to say the coin is broken.

Solve the problem correctly. Show all of your work.

No, the coin is not necessarily broken

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s rolled 60 times, and 6 comes up 12 times. What is the experimental probability of rolling a 6? Student's ans

Divide the number of 6s by the total rolls.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the count. The probability is 12/60 = 0.2.	**0.2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

is tossed 10 times and lands heads 3 times. What is the experimental probability of heads? Student's answ

Divide heads by total tosses.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the theoretical probability. Experimental probability is 3/10 = 0.3.	**0.3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

etball player makes 7 of 10 free throws. Estimate the probability she makes her next free throw. Student's an

Divide made shots by total shots.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the number made. The estimate is 7/10 = 0.7.	**0.7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A fair six-sided die is rolled. What is the probability of rolling a 4? Student's answer: 1/3

Assume each outcome is equally likely.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 6 equally likely outcomes, so P(4) = 1/6.	**1/6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

l gives outcomes A, B, C, and D with probabilities 0.2, 0.3, and 0.4 for A, B, and C. What is P(D)? Student's ans

The probabilities must add to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
0.2 + 0.3 + 0.4 = 0.9. P(D) = 1 - 0.9 = 0.1.	**0.1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A spinner has 8 equal sections, and 3 are blue. What is P(blue)? Student's answer: 3/5

Count the blue sections out of all sections.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The total is 8 sections, not 5. P(blue) = 3/8.	**3/8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bag has 12 marbles, 4 of them red. A marble is picked at random. What is P(red)? Student's answer: 4/8

Count the red marbles out of all marbles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The total is 12 marbles, not 8. P(red) = 4/12 = 1/3.	**1/3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

inner has 4 equal sections with probabilities 0.2, 0.2, 0.2, and 0.2. Student says: This is a valid probability mo

Check whether the probabilities add to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The probabilities add to 0.8, not 1. This is not a valid model.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s chosen at random from the 26 letters of the alphabet. There are 5 vowels. What is P(vowel)? Student's ans

Count the vowels out of all letters.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The total number of letters is 26, not 21. P(vowel) = 5/26.	**5/26**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two fair coins are flipped. What is the probability of getting two heads? Student's answer: 1/2

List all possible outcomes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The outcomes are HH, HT, TH, and TT. Only HH works, so P = 1/4.	**1/4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two fair dice are rolled. What is the probability that the sum is 7? Student's answer: 1/12

List the outcomes that make 7.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
There are 6 pairs that make 7 out of 36 total outcomes. 6/36 = 1/6.	**1/6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A coin is flipped and a die is rolled. How many total outcomes are there? Student's answer: 8

Multiply the number of outcomes for each event.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The coin has 2 outcomes and the die has 6. The total is 2 × 6 = 12.	**12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 red and 2 blue marbles. Two marbles are picked without replacement. What is P(both red)? Student's an

Use the marbles left after the first pick.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student replaced the first marble, but the problem says not to. P = 3/5 × 2/4 = 3/10.	**3/10**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A student has 2 shirts and 3 pairs of pants. How many different outfits can she make? Student's answer: 5

Multiply the number of choices.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the choices. Each shirt can go with each pair of pants: 2 × 3 = 6 outfits.	**6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two coins are flipped. What is the probability of getting at least one head? Student's answer: 1/2

Use the outcomes or the complement.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Only TT has no heads. P(at least one head) = 1 - 1/4 = 3/4.	**3/4**

Original task. Format inspired by error analysis routines.